

National Examination Paper — Copy B

[Centre 0028 | Hall 001 | Print #0001] — CONFIDENTIAL

- Q1.** Explain the working principle of a digital watermark and its application in forensic document authentication systems that must survive physical print and photograph pipelines.
- Q2.** A train departs from Station A at 60 km/h and returns at 40 km/h. Calculate the average speed over the entire journey and justify why arithmetic mean of speeds is incorrect.
- Q3.** Describe how Reed-Solomon error correction codes recover corrupted data. Explain the relationship between the number of parity symbols and the number of correctable errors.
- Q4.** Define homography in computer vision and show how perspective transformation is applied to normalize a document photographed from an angle.
- Q5.** Explain symmetric versus asymmetric encryption with one real-world application for each. Discuss the computational cost trade-offs and why hybrid schemes are commonly used.
- Q6.** Using integration, find the area enclosed between y equals x squared and y equals x plus two, showing all steps and verifying the result geometrically.
- Q7.** Describe Shamir's Secret Sharing scheme and explain why a threshold k -of- n approach is more robust than splitting a key into n non-overlapping equal parts.
- Q8.** Analyze the cryptographic properties of secure hash functions including pre-image resistance and collision resistance in digital forensics.
- Q9.** Explain the differences between physical and digital watermarking in the context of high stakes examination question paper custody.
- Q10.** Derive the error correction capability formula for a Reed-Solomon code with n total symbols and k message symbols.